POLITEKNIK MALAYSIA PORT DICKSON	**COURSE OUTLINE**
	DEPARTMENT OF CIVIL ENGINEERING
	EFFECTIVE DATE: 13 JANUARY 2025
	VERSION: *06062024_1_Effective : Session_1_2024/2025*

1.	A. NAME OF COURSE	CONSTRUCTION MATERIALS	
	B. COURSE CODE	DCC10292	
	C. PROGRAM	DIPLOMA IN CIVIL ENGINEERING	
2.	SYNOPSIS	**CONSTRUCTION MATERIALS** course aims to provide students with a thorough understanding of construction materials in civil engineering. The curriculum offers a comprehensive overview of civil engineering materials, focusing on concrete materials, including cement, aggregate, water and admixtures. It extensively covers concrete mixtures and the properties of both fresh and hardened concrete. Additionally, the course explores the properties of timber, types and characteristics of bricks and concrete blocks, and the basic concepts of steel materials. It also includes knowledge about advanced construction materials and methods related to each topic.	
3.	NAME(S) OF ACADEMIC STAFF	UMI HANI BINTI ABDUL RAHMAN	
4.	SEMESTER AND YEAR OFFERED	Semester 1 / Year 1	
5.	CREDIT VALUE	2	
6.	PREREQUISITE/ CO-REQUISITE (IF ANY)	None	
7.	A. PROGRAMME EDUCATIONAL OBJECTIVES (PEO): The Diploma in Civil Engineering programme shall produce semi-professionals who are:		
	PEO 1	Working in the field of civil engineering	/
	PEO 2	Lead or a team member to support their role in industries	/
	PEO 3	Engaged in activities to enhance knowledge or starting/embark their own enterprise	
	PEO 4	Fulfill professional and communities' responsibilities, conforming to ethical and environmental values	/
	B. PROGRAMME LEARNING OUTCOMES (PLO): Upon completion of this programme, students should be able to:		
	PLO 1	Apply knowledge of applied mathematics, applied science, engineering fundamentals and an engineering specialization as specified in DK1 to DK4 respectively to wide practical procedures and practices.	
	PLO 6	When solving well-defined engineering problems, evaluate sustainable development impacts* to: society, the economy, sustainability, health and safety, legal frameworks, and the environment. (DK1, DK5, and DK7)	
	PLO 9	Communicate effectively and inclusively on well-defined engineering activities with the engineering community and with society at large, by being able to comprehend the work of others, document their own work, and give and receive clear instructions.	

C. COURSE LEARNING OUTCOMES (CLO):

Upon completion of this course, students should be able to:

CLO 1	Explain civil engineering materials and method in construction practice. (C3, PLO1)
CLO 2	Propose selection of material and method in construction practice with consideration of construction specification, condition and sustainability. (A3, PLO6)
CLO 3	Present the task given with effective communication skills. (A2, PLO9)

8. TRANSFERABLE SKILLS — None

9. DISTRIBUTION OF STUDENT LEARNING TIME (SLT)

COURSE CONTENT OUTLINE	CLO*	TEACHING AND LEARNING ACTIVITIES L	P	T	O	GL	IL	SLT
1.0 Introduction to Construction Materials								
1.1 Explain the overview of construction material.	CLO 1	1					1	2
1.2 Describe overview of construction method in civil engineering project.	CLO 1	1					1	2
1.3 Describe the building construction method in civil engineering.	CLO 2	2					2	4
	CLO 3	0.5					0.5	1
2.0 Concrete Materials and Mixture								
2.1 Explain the materials in concrete.	CLO 1	1					1	2
2.2 Explain the basic knowledge of concrete mixture.	CLO 1	1					1	2
2.3 Utilize the basic knowledge of concrete mixture to classify fresh concrete and hardened concrete.	CLO 1	1					1	2
2.4 Explain concrete application as advanced construction materials.	CLO 2	3					3	6
	CLO 3	0.5					0.5	1
3.0 Timber								
3.1 Explain properties of local timber.	CLO 1	1					1	2
3.2 Explain process of timber production.	CLO 1	1					1	2
3.3 Explain timber product.	CLO 2	2					2	4
	CLO 3	0.5					0.5	1
4.0 Brick								
4.1 Explain about brick and block.	CLO 1	1					1	2
4.2 Explain manufacturing of brick.	CLO 2	2					2	4
	CLO 3	0.5					0.5	1
4.3 Explain about brickwork.	CLO 1	1					1	2
5.0 Steel								
5.1 Explain the basic concept of iron and steel.	CLO 1	2					2	4
5.2 Explain the application of steel in construction.	CLO 2	1					1	2
	CLO 3	1					1	2
TOTAL		**24**	**0**	**0**	**0**	**0**	**24**	**48**

CONTINUOUS ASSESSMENT	%	F2F	NF2F	SLT
Test	20	1	3	4
Mini Project Progress	35	2	10	12
Mini Project Presentation	15	1	5	6
Mini Project Report	30	0	10	10
TOTAL	**100**	**4**	**28**	**32**

FINAL ASSESSMENT	%	DURATION	NF2F	SLT
		0	0	0

GRAND TOTAL	80

Please tick (√) if this course is an Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT) ☐

Note:
*Indicate the CLO based on the CLO's numbering in Item 7.
L = Lecture, P= Practical, T = Tutorial, O= Others, F2F=Face to Face, NF2F=Non Face to Face, GL-Guided Learning, IL=Independent Learning

10.	SPECIAL REQUIREMENT OF RESOURCES TO DELIVER THE COURSE	None
11.	REFERENCES	**Main reference supporting the course:** Mamlouk, M. S. and Zaniewiski, J. P. (2021). Materials for Civil and Construction Engineers. Fourth Edition. United Kingdom: Pearson. (ISBN: 9780137505586) **Additional references supporting the course:** Lee, L. S. and Estrada, H (2020). Materials for Civil Engineering: Properties and Applications in Infrastructure. First Edition. New York: McGraw-Hill Education. (ISBN: 9781259862618) Standard Specifications for Building Works. (2020). Jabatan Kerja Raya Malaysia. (ISBN:978-967-2284-14-7)
12.	OTHER ADDITIONAL INFORMATION	None

13. TEACHING SCHEDULE:

Topic No.	Topic/Content	Recommended Contact Hours	Assessment Method	Week
1.0	**INTRODUCTION TO CONSTRUCTION MATERIALS** This topic included the overview of construction materials, their definition, role, and classification by source, composition, sustainability, and usage. It introduced the construction methods, including steps and stages in civil engineering projects. Discuss building construction methods with sustainability impacts and differentiate between conventional and advanced methods.	4.5 hours Lecture	• Mini project report	W1 – W3
2.0	**CONCRETE MATERIALS AND MIXTURE** This topic covers the concrete materials: cement, aggregates, water, and their roles. It discussed concrete mixtures, proportions, admixtures, fresh and hardened concrete properties, workability, and tests. It also explained advanced concrete types and their sustainability aspects.	6.5 hours Lecture	• Test • Mini project report • Mini project progress 1	W3 – W6
3.0	**TIMBER** This topic covers the local timber properties, types, characteristics, and defects. It explained timber production processes and discuss varieties of timber products.	4.5 hours Lecture	• Test • Mini project report • Mini project progress 1	W7 – W9

	4.0	**BRICK** This topic introduces bricks and blocks: types, properties, and applications. It described brick manufacturing process and discuss advanced brick applications, like lightweight blocks. It also explained several types of brickwork bonds.	4.5 hours Lecture	• Test • Mini project report • Mini project progress 2	W9 - W11
	5.0	**STEEL** This topic introduces the students to the iron and steel: properties, advantages, and disadvantages of structural steel. It explained steel applications in construction, including modular and prefabricated construction, reinforcement, roofing and cladding, and building frameworks**.**	4 hours Lecture	• Mini project report • Mini project progress 2	W12 – W14

14. ASSESSMENT

The course assessment consists of:

i. Continuous Assessment (CA) – 100%
ii. Final Examination (FE) – 0%

Assessment	Quantity	Percentage (%)
Test	1	20%
Mini project progress	2	35%
Mini project report	1	30%
Mini project presentation	1	15%

Prepared by;

.....

(Name: UMI HANI BINTI ABDUL RAHMAN)
COURSE COORDINATOR
Date: 8/1/2025

Verified by;

.....

(Name: NOR HANIZA BINTI MUSTAFAR KAMAR)
HEAD OF DEPARTMENT/HEAD OF PROGRAMME/
HEAD OF COURSE
Date: 12/01/2025

NOR HANIZA BINTI MUSTAFAR KAMAR
Ketua Program
Diploma Kejuruteraan Awam
Jabatan Kejuruteraan Awam
Politeknik Port Dickson